SONiX 32-Bit Cortex-M0 Micro-Controller

DMA UARTtoSPI Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/29	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of DMA peripheral(UART) to peripheral(SPI) mode.

2 FEATURE

- DMA UART to SPI mode.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect UART0 TX to UART0 RX, SPI0 MISO to SPI0 MOSI.

4 EXAMPLE CODE

4.1 Initialization

Initial UART, SPI and DMA function. User can modify the settings of UART, SPI and DMA in these function.

```
// Initial UART & SPI
UART0_Init();
SPI0_Init();

//Auto-SEL enable
SN_SPI0->CTRL0_b.SELDIS = SPI_SELDIS_EN;

// Disable UART ISR when using DMA UART & SPI
NVIC_DisableIRQ(UART0_IRQn);
NVIC_DisableIRQ(SPI0_IRQn);

// Init DMA
DMAUartToSpiInit();
// DMA SPI RX to SRAM for data check
DMASpiToSramInit();
```

UART will transmit 32 bytes of data from 0x00 to 0x1F.

```
for(i = 0; i<DMA_TEST_SIZE ;i++)
{
  // UART0 TX data
  UART0_SendByte(i);
}
```

4.2 DMA function enable

After DMA function enable. DMA request will be issued when the SPI TX FIFO is less than the threshold and UART RX is not empty.

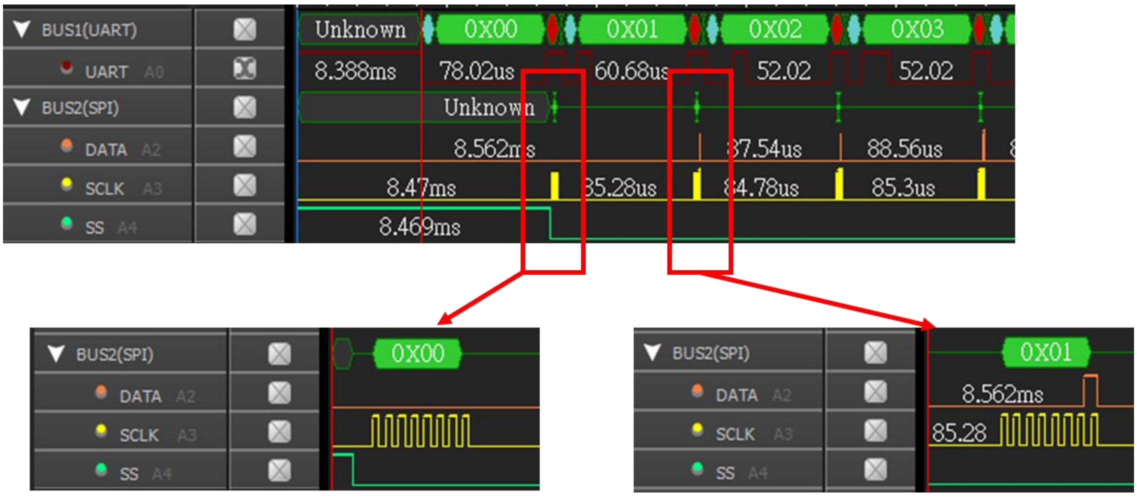
```
// Start DMA UART TX transfer & DMA SPI
UART0_DMASizeSet(DMA_TEST_SIZE);
UART0_DMAEnable();
SPI0_DMAEnable(mskSPI_TX_DMA_EN|mskSPI_RX_DMA_EN, DMA_TEST_SIZE);
DMASpiToSramStart();
DMAUartToSpiStart();
```

The DMA TC flag and interrupt will be set when all data transfers are completed.

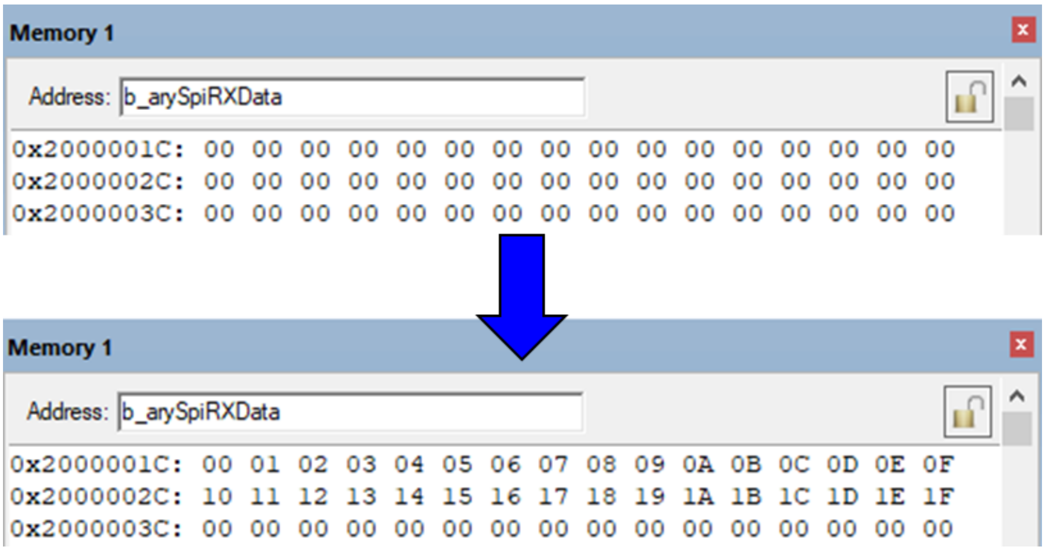
```
// Wait for DMA TC interrupt flag
while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
while(stDMA0IntFlag[eDMA_CH1].Flag.bits.TC != 1);
```

5 RESULT

The data sent or received can be observed on UART and SPI bus.



The results can be observed through Watch window or Memory window.



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